

Deshan Lokuge

NETWORK | SECURITY | IoT ENGINEERING

📍 Baddegama, Galle, Sri Lanka

☎ +94 76 366 5681 ✉ deshanlokuge124@gmail.com 🌐 github.com/Deshan in linkedin.com/Deshan

Summary

Electrical and Information Engineering undergraduate specializing in computer networking, telecommunications and security with hands-on experience in enterprise network operations and IoT system development. Built wireless security and IoT monitoring solutions by combining networking with embedded hardware.

Education

○ BSc. Eng (Hons.) Electrical and Information Engineering 2022 – Present

University of Ruhuna

GPA: 3.60

Focus: Networking, Telecommunications and Security | Minor: Electrical Engineering

Selected Coursework: Telecommunication and Computer Networking; Operating Systems; Information Security; Data Networks and Design; Embedded Systems Design

○ G.C.E. Advanced Level Examination

2017 – 2020

Richmond College, Galle

Z-Score: 1.9284

Experience

Internship — Samson International PLC | 🌐 Repo | Certificate

Aug 2025 - Oct 2025

- Designed and deployed a production-monitoring network across press machines, replacing paper-based counts with real-time ThingsBoard web/mobile dashboards and remote configuration.
- Improved IoT device security by using unique access tokens and separate Wi-Fi and OTA credentials, while adding automatic MQTT reconnection for reliable communication.
- Designed scalable PCBs and improved fault tolerance by isolating sensor inputs and adding signal filtering and debounce logic.

Internship — SLT Telecom PLC | 🌐 Repo | Certificate

May 2024 - Aug 2024

- Supported enterprise network monitoring and maintenance at SLT OPMC – Galle, assisting with fault identification and basic troubleshooting across operational telecom infrastructure.
- Contributed to the Fazenda Smart Agro project at SLT Digital Lab, collaborating with engineers on planning, prototype testing, technical documentation, and system refinement.
- Integrated a microcontroller-based smart irrigation system with ThingsBoard dashboards, calibrated sensors and tested real-time monitoring, alerts and remote control.

Technical Skills

Networking & Security: Cisco, GNS3, Packet Tracer, Wireshark, VLANs, Zabbix, Zero-Trust Security

Programming & Automation: Python, C/C++, Bash, Ansible, SQL/MySQL, Git/GitHub, React.js

Telecom & Systems: Open-RAN, Open5GS, srsRAN, Docker, Kubernetes, Linux/Ubuntu

IoT & Embedded: Arduino, MQTT, ThingsBoard, PCB design, EasyEDA, Proteus, Microchip Studio

Software & Tools: VS Code, Jira, MATLAB, Figma, Grafana

Projects

1. **Threat Containment Strategies for Compromised xApps in Open RAN** | 🌐 Repo

- Developed **ZT-XGuard**, a zero-trust runtime security framework with workload attestation and deployed a Kubernetes testbed integrating OSC Near-RT RIC, Open5GS, srsRAN and operational xApps.
- Designed and validated a statistical **MEWMA/Hotelling- T^2** resource-anomaly detector, achieving a **98.07%** overall detection rate across evaluated resource-exhaustion attacks.

- Built a runtime system-call intrusion-detection system using **Falco** to detect credential/identity tampering, unauthorized shell execution and lateral-movement attempts between xApps.
- Diagnosed and resolved xApp E2 node connectivity in **NIST's O-RAN Testbed Automation** tool; documented the fix in **#Issue** and received acknowledgement from the project builders.
- Integrated a multi-layer automated containment using Kubernetes API, RBAC, **Calico CNI** and **SPIFFE/SPIRE** identity revocation to isolate compromised xApps from Near-RT RIC, achieving **75.4 ms** containment latency.

2. **Department-Level Campus Network Design & Management** | Repo

- Designed and implemented a scalable **enterprise campus network** using GNS3, Cisco IOS, incorporating VLANs, OSPF, ACLs and security management.
- **Automated** network provisioning and configuration management using Python and Ansible across Cisco routers and switches.
- Implemented centralized **network monitoring** with **Zabbix** for real-time device monitoring, fault detection and performance analysis.

3. **Offensive and Defensive PEN Tool** | Repo

- Designed and implemented a portable wireless penetration testing tool for network security assessments. Features included **Wi-Fi/Bluetooth** scanning, **MAC sniffing**, **Evil Twin** and **Portal attacks**, packet monitoring, and deauthentication attack detection.
- Applied penetration testing methods and wireless security protocols to perform offensive and defensive security evaluations focused on **PCAP analysis** and traffic monitoring.

4. **Secure xApp Onboarding** | Repo

- Designed an onboarding pipeline that verifies **GPG-signed** xApp descriptors and **Cosign-signed** container images before deployment to the Near-RT RIC in O-RAN.
- Integrated SPIRE/SPIFFE workload identity with **short-lived SVID** issuance to bind verified software to authorized Kubernetes workloads and reject tampered xApps.

5. **PayFlow - Online Voucher Application Portal** | Repo

- Developed a web-based application to streamline reimbursement and voucher requests, improving financial management for university academic and non-academic staff.
- Acted as **Scrum Master**, managing Jira boards, and contributed to frontend development, including **Clerk-based authenticated user logins** using React.js.

Professional Certifications and Licenses

- Cisco Networking Academy — Cisco Certified Network Associate: ITN  | SRWE* | ENSA*
- Computer Networks - Huawei  ◦ Endpoint Security - Cisco 
- Introduction to Cybersecurity - Cisco  ◦ Java and Python Programming 

Volunteering and Co - Curricular Involvement

- Volunteered for the Mehewara project organized by second-year undergraduates of the faculty, aimed at supporting O/L students.
- Volunteer member of the ECSL Electrician's Guide team, University of Ruhuna Team.
- Haxtreme 2.0 - Coding Competition - **2023**.
- Membership - Institution of Engineers, Sri Lanka (IESL) - Membership number: S-32500

References

Dr. Chatura Seneviratne
Head of Department, DEIE
Faculty of Engineering, University of Ruhuna
Email: chatura@eie.ruh.ac.lk

Eng. Ilumini Muthunaike
Engineer-Research & Development
Digital Lab, Sri Lanka Telecom PLC
Email: ilumini@slt.com.lk